SVM & Logistic Regression for Wine-Quality Prediction: A Machine Learning Project

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February 27, 2026

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1 Introduction

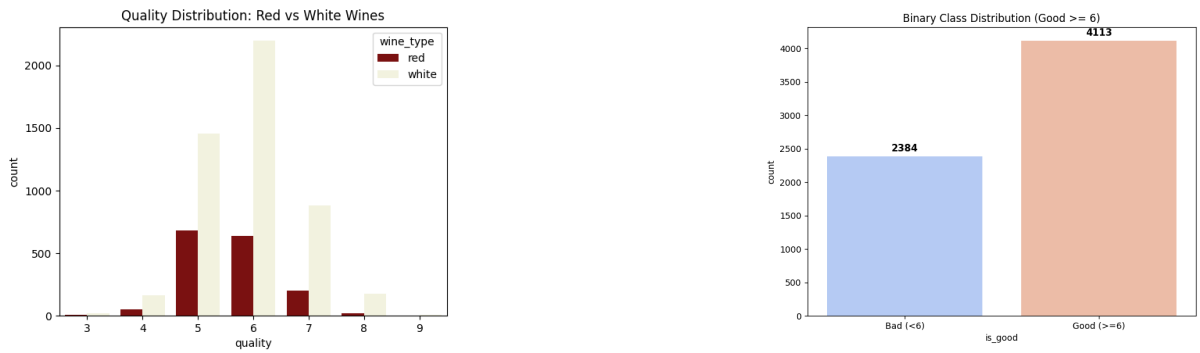
This report presents the implementation and analysis of two fundamental binary classification algorithms: Support Vector Machine (SVM) and Logistic Regression (LR). The primary objective of this project is to implement both models from scratch and apply them to a binary classification task: predict whether a wine can be considered "good" or "bad", considering both red and white wine samples simultaneously.
The analysis follows a standard machine learning approach. After an initial data exploration and preprocessing phase, the models are trained, evaluated, and the results compared. Hyperparameter tuning and cross-validation procedures are implemented in order to ensure proper model generalization and robustness of the results.
In particular, both models were first implemented in the standard linear form and then applied to the dataset; sub-sequently, the models were extended to a kernel formulation using two non linear functions: polynomial kernel and gaussian kernel.

2 Dataset

The dataset used in this project is obtained by merging the red and white wine datasets available in the Wine Quality Dataset. Link:
The combined data set contains 6,497 samples. Each observation is described by 11 continuous input variables based on physicochemical laboratory tests. The target variable, *quality*, is an integer score ranging between 0 and 10 and it is based on sensory data.
The input variables are:

- fixed acidity
- volatile acidity
- citric acid
- residual sugar
- chlorides
- free sulfur dioxide
- total sulfur dioxide
- density
- pH
- sulfates
- alcohol

In order to reformulate the problem as a binary classification task, the original quality score was transformed into a binary variable. Wines with a quality score greater than or equal to 6 were assigned to Class 1 ("good" wines), while wines with a score lower than 6 were assigned to Class 0 ("bad" wines).
After binarization, the dataset exhibits a moderately imbalanced class distribution, with a higher proportion of good wines compared to bad ones. Additionally, the dataset is imbalanced with respect to the wine type, with a larger proportion of white wines compared to red wines. This imbalance was taken into account during the train-test splitting procedure.



(a) Original quality distribution

(b) Binary class distribution

Figure 1: Class distributions before preprocessing.

3 Data Exploration and Preprocessing

An initial exploratory data analysis was conducted to better understand the structure and characteristics of the dataset. A preliminary inspection confirmed the absence of missing values. Summary statistics, including mean, standard deviation, minimum and maximum values, were computed for all numerical variables.
Histograms and boxplots were plotted for each feature in order to analyze their distributions and detect potential outliers that could interfere with the analysis.

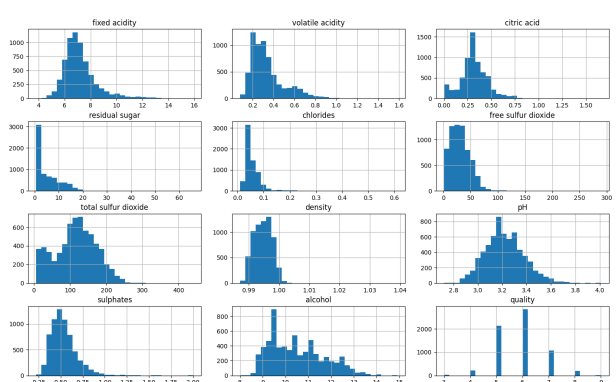


Figure 2: Features histograms

Several variables exhibited the presence of extreme values, particularly residual sugar, chlorides, free sulfur dioxide, and total sulfur dioxide. To reduce the impact of extreme observations, outliers were removed using the Interquartile Range (IQR) method. Specifically, observations falling outside the interval

$$[Q_1 - 1.5 \cdot IQR,\ Q_3 + 1.5 \cdot IQR]$$

were removed.

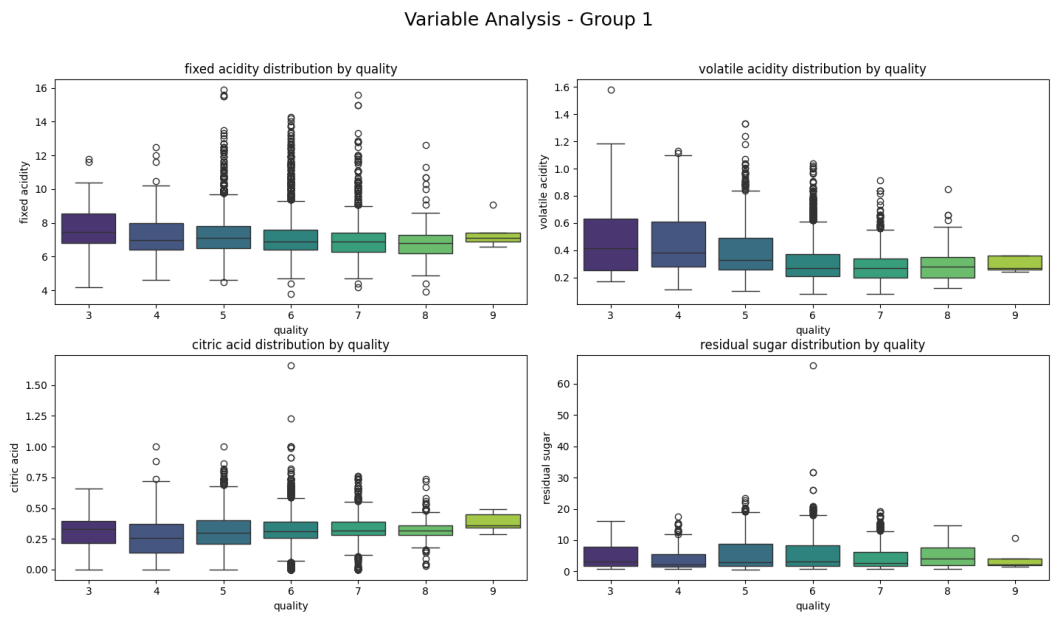
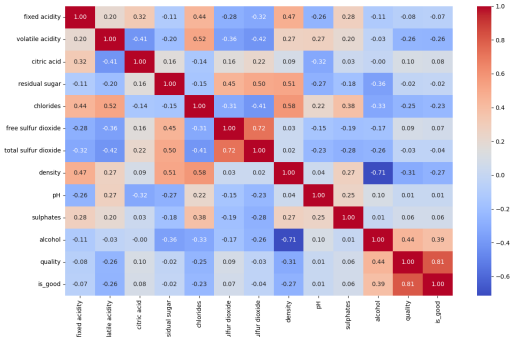


Figure 3: Outliers

As a result, the total number of samples was reduced from 6,497 to 6,026. The class distribution was re-evaluated after this procedure to ensure that no significant distortion was introduced.

A correlation analysis was then performed to study the linear relationships between features and the target variable. The results show that alcohol has the strongest positive correlation with quality (approximately 0.44), indicating that wines with higher alcohol content tend to receive higher quality scores. In contrast, volatile acidity and density show negative correlations with quality. Moreover, strong multicollinearity is observed between some features.



(a) Confusion Matrix

(b) Pairplot

As we can see from these scatterplots, although some linear trends are observable, the separation between classes is not perfectly linear, motivating the later introduction of kernel methods.
Finally, a stratified 80/20 train-test split was performed in order to preserve the original proportion of wine types in both training and test sets. Feature standardization, an essential step for the correct implementation of the models, was applied exclusively to the training set, and the same transformation was subsequently applied to the test set.

4 Linear models: SVM and Logistic Regression

After preprocessing and feature standardization, two linear classifiers were implemented from scratch: Support Vector Machine (SVM) and Logistic Regression (LR).
Both linear SVM and Logistic Regression were implemented from scratch following the theoretical formulations provided in the course lecture notes. The models were optimized using stochastic gradient descent (SGD) with ℓ_2 regularization, which are types of OGD algorithms that run over a sequence of examples randomly drawn from the training set. In particular, the SVM minimizes the regularized hinge loss, while Logistic Regression minimizes the regularized logistic loss.

4.1 Label Encoding

Since both SVM and Logistic Regression were implemented using surrogate loss functions defined for $y \in \{-1, +1\}$, the original binary labels $\{0, 1\}$ were converted into:

$$y = \begin{cases} +1 & \text{if quality } \geq 6 \\ -1 & \text{otherwise} \end{cases}$$

4.2 Support Vector Machine (SVM)

Support Vector Machine, an algorithm for learning linear classifiers, was implemented in its soft-margin formulation. Given a training set, the objective function minimized is:

$$F(w) = \frac{1}{m}\sum_{i=1}^{m}\left[1 - y_i w^\top x_i\right]_+ + \frac{\lambda}{2}\|w\|^2 \tag{1}$$

where $[z]_+ = \max(0, z)$ denotes the hinge loss and $\lambda > 0$ is the regularization parameter. The gradient of the regularized hinge loss is given by:

$$\nabla \ell_i(w) = \begin{cases} -y_i x_i + \lambda w & \text{if } y_i w^\top x_i < 1 \\ \lambda w & \text{otherwise} \end{cases} \tag{2}$$

The parameters were updated using stochastic gradient descent:

$$w_{t+1} = w_t - \eta \nabla \ell_i(w_t) \tag{3}$$

where η denotes the learning rate.

4.3 Logistic Regression

Logistic Regression is an algorithm for learning linear classifiers in a binary classification problem where the output is a probability, instead of a binary prediction and it was implemented using the logistic loss function. The linear predictor is defined as:

$$g(x) = w^\top x \tag{4}$$

and the logistic function is:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{5}$$

The regularized logistic loss minimized is:

$$F(w) = \frac{1}{m}\sum_{i=1}^{m}\log\left(1 + e^{-y_i w^\top x_i}\right) + \frac{\lambda}{2}\|w\|^2 \tag{6}$$

The gradient of the loss is:

$$\nabla \ell_i(w) = -\sigma(-y_i w^\top x_i) y_i x_i + \lambda w \tag{7}$$

The weight update rule is:

$$w_{t+1} = w_t - \eta \nabla \ell_i(w_t) \tag{8}$$

4.4 Hyperparameter tuning and model selection

For both models, hyperparameter tuning was implemented using a 5-fold cross-validation to ensure the use of the best parameters during the final implementation. The following parameters were optimized:

- Learning rate
- Regularization parameter λ
- Number of training epochs

The best combination of parameters was selected based on the cross-validation accuracy on the whole training set.

4.5 Model evaluation

Both models were finally evaluated on the independent test set using the following performance metrics:

- Accuracy
- Precision
- Recall
- F1-score

5 Kernel methods

Linear classifiers may suffer from high bias when the data are not linearly separable in the feature space. From the exploratory data analysis, several pairwise feature plots revealed a significant overlap between the two classes and suggested the presence of nonlinear relationships among variables. Although linear models provide a reasonable baseline, these observations motivate the use of more flexible classifiers. Two kernel methods were implemented. This approach allows us to map the input data into a higher dimensional feature space without explicitly computing the transformation.

5.1 Polynomial kernel

The polynomial kernel of degree n is defined as:

$$K_n(x, x') = (1 + x^\top x')^n \tag{9}$$

This kernel corresponds to an implicit feature expansion including all monomials up to degree n. Increasing the degree increases model flexibility, but may also increase the risk of overfitting.

5.2 Gaussian (RBF) Kernel

The Gaussian kernel is defined as:

$$K_\gamma(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\gamma}\right) \tag{10}$$

The parameter γ controls the width of the Gaussian functions. Small values of γ produce highly localized decision boundaries (higher variance), while large values of γ result in smoother decision boundaries (higher bias).

5.3 Kernel SVM and Logistic Regression

The already implemented models were extended to their kernel implementation by replacing inner products $x^\top x'$ with the kernel function $K(x, x')$. The performance of Kernel SVM and Logistic Regression strongly depends on the choice of some specific hyperparameter: the polynomial kernel requires selecting the degree n, while the Gaussian kernel depends on the parameter γ.
In particular, during the hyperparameter tuning, some parameters as the number of epochs were fixed to smaller values than the ones used in the final training for computational reasons (the time of training) specifically for the LR both in the polynomial and gaussian kernel. To ensure the selection of the best combination, still a 3-folds cross-validation was implemented and the combination achieving the highest accuracy was selected.
The final models were then evaluated on the independent test set, which was never used during the tuning, in order to obtain an unbiased estimate of generalization performance.

6 Evaluation and Analysis

6.1 Performance Metrics

The performance of all the six models was evaluated using accuracy, precision, recall and F1-score: *accuracy* measures the overall proportion of correctly classified samples, while *precision* and *recall* provide insight into the trade-off between false positives and false negatives. The *F1-score* summarizes this balance.
As shown in Figure 5, the Gaussian Logistic Regression model achieves the highest accuracy (0.831), followed by the SVM Gaussian and SVM linear.
The polynomial SVM model exhibits extremely high recall but relatively lower accuracy. This behavior indicates that the model tends to classify almost all samples as belonging to the positive class, resulting in many false positives.
Overall, the Gaussian kernel improves performance compared to linear models, confirming that introducing nonlinearity provides a better representation of the decision boundary.

Performance Metrics Comparison
Best Model: LR Gaussian (Accuracy = 0.831)

model	accuracy	precision	recall	f1
LR Linear	0.715	0.844	0.699	0.765
SVM Linear	0.769	0.802	0.864	0.832
LR Polynomial	0.753	0.784	0.866	0.823
SVM Polynomial	0.705	0.695	0.989	0.816
LR Gaussian	**0.831**	**0.857**	**0.895**	**0.875**
SVM Gaussian	0.772	0.839	0.812	0.825

Figure 5: Performance metrics comparison.

Although accuracy improves with non linear kernels, it is fundamental to analyze the confusion matrices in order to understand the type of classification errors made by each model.

6.2 Confusion Matrix Analysis

The confusion matrix (Figure 6), reveals significant difference among models in the errors made.

- **Linear models** show a relatively balanced distribution of false positives and false negatives. This suggests moderate bias but stable generalization behavior.
- **SVM with polynomial kernel** produces a very high number of false positives. This indicates that the model tends to classify most samples as positive, which is a typical symptom of overfitting and excessive model complexity.
- **Gaussian Logistic Regression** achieves a better balance between sensitivity and specificity. It reduces both false positives and false negatives compared to the other models, resulting in the best overall trade-off.
- **Gaussian SVM** also improves over the linear version, but it does not achieve the same balance as the Gaussian Logistic Regression model.

It is relevant to underline that the Gaussian Logistic Regression model produced the lowest number (203) of classification errors over all the others.

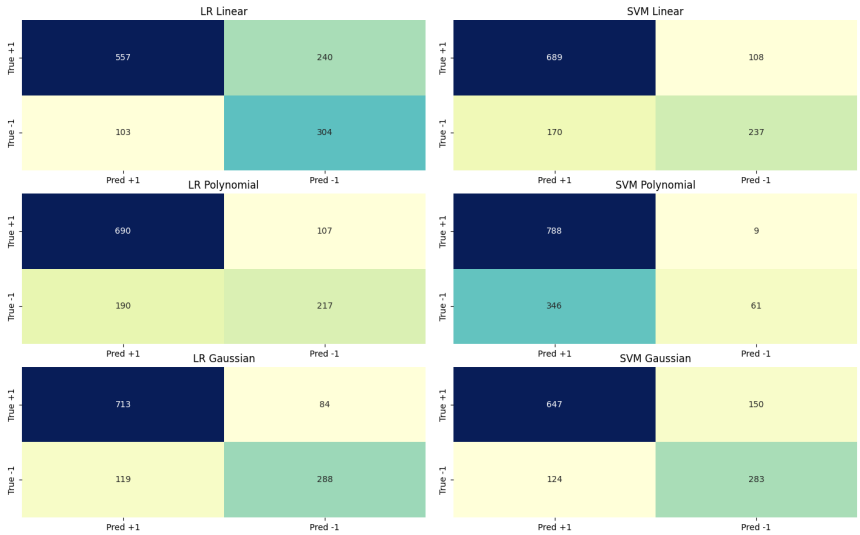


Figure 6: Confusion Matrix

6.3 ROC Curve Analysis

ROC Curves (Figure 7) provide an evaluation on models performance. The Gaussian models achieve the highest Area Under the Curve (AUC), confirming their superior performance.

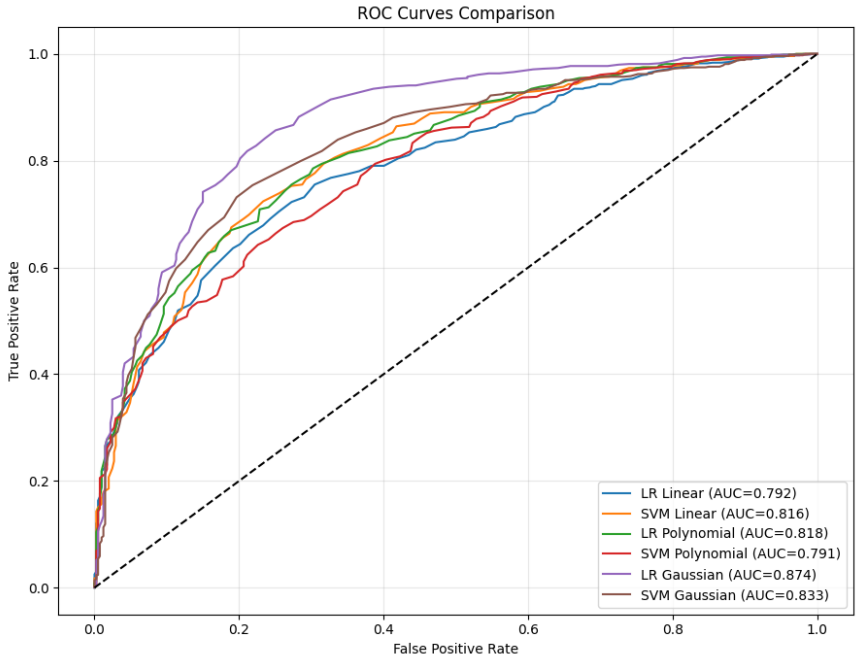


Figure 7: ROC Curve

In particular, the LR Gaussian model achieves the highest AUC value (0.874), indicating better ranking ability between positive and negative classes. Linear models show lower AUC values, suggesting that their decision boundary is less flexible in capturing complex patterns.

6.4 Learning Curves and Model Complexity

The learning curves (Figure 8) are a graphical representation that shows how the performance of the model on a set changes with the number of training iterations or the amount of training data and provide insight into the bias-variance behavior of the models:

- **Linear models** exhibit stable convergence with a small gap between training and validation loss. This behavior suggests good generalization capability and relatively low variance.
- **SVM with polynomial kernel** shows unstable behavior and extremely large loss values during the initial training epochs.
- **SVM with gaussian kernel** still requires many epochs because the optimization problem is more complex than in the linear case.
- **Gaussian Logistic Regression** achieves the best balance between flexibility and generalization performance, combining stable convergence with strong predictive accuracy.

The initial spike observed in the training loss of the kernel SVM models can be explained by the learning rate schedule adopted during training:

$$\eta_t = \frac{1}{\lambda t}.$$

During the first iteration, when $t = 1$, the learning rate becomes $\eta_1 = \frac{1}{\lambda}$ which can be very large, as in this case, if the regularization term is small (lambdas = [1e-6, 1e-5, 1e-4] for Polynomial, lambdas = [0.001, 0.01] for Gaussian). This produces a very large initial value of the coefficients α_j.
In the kernel formulation, the decision function is

$$f(x) = \sum_{j=1}^{m} \alpha_j y_j K(x_j, x),$$

so, the consequence of a high value for α_j is the increase in the magnitude of $f(x)$ and to different variations in the hinge loss term $(1 - y_i f(x_i))_+$ and consequentially in the total loss.

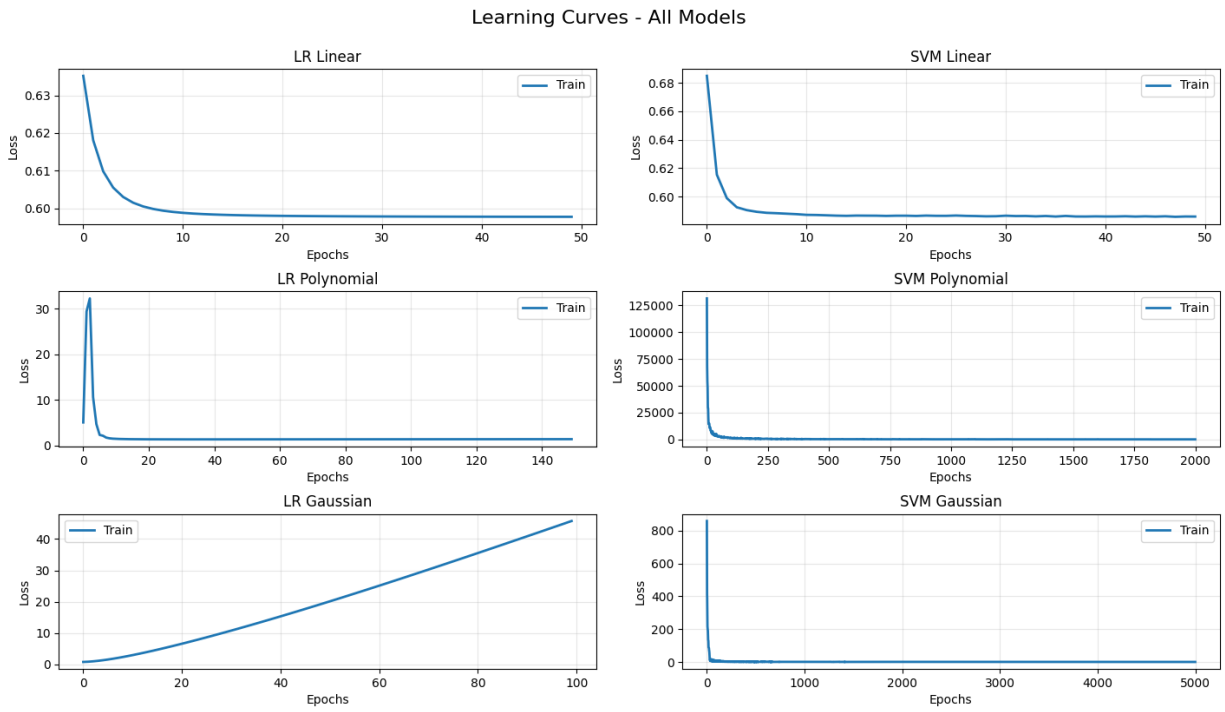


Figure 8: Learning Curves

7 Conclusion

Although nonlinear kernel models improve performance compared to linear versions, which are limited by their inability to capture nonlinear relationships among features, none of the models achieves extremely high accuracy. This outcome reflects the difficulty of the wine quality classification task, where the two classes may not be perfectly separable, and where some overlap between good and bad wines may be present. After the analysis, it is possible to affirm that:

- **Linear models** introduce a too simple linear relation between the features, which is not sufficient to correctly capture the real dataset characteristics.
- **Polynomial kernels** introduce a too high flexibility, leading in some cases to overfitting.
- **Gaussian Logistic Regression model** achieves the best trade-off between flexibility and generalization, ensuring the best compromise between accuracy, stability and robustness among the tested approaches.

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